OMEGA SERIES 4.2

A Disformal Chameleon Effective Field Theory

with Internal Sector Structure for the Dark Energy Scalar

Disformal kinetic loophole · Three-regime sound-speed map · The δ = 0.00017 off-principle · PPN-safe · GW170817-compliant · IAXO target: $g_{\phi\gamma} \sim 10^{-11}$/GeV

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ABSTRACT

We construct a disformal scalar-tensor effective field theory with a chameleon potential and internal sector structure for the dark energy scalar ϕ. The action contains a standard canonical kinetic term, a Ratra-Peebles inverse-power potential, a conformal coupling to the electromagnetic sector, and a higher-derivative disformal term $(\partial\phi)^4/M^4$ that implements a kinetic loophole: the scalar is screened in static high-density environments (satisfying MICROSCOPE and Cassini bounds by 16 orders of magnitude) but activates cosmologically where ϕ-dot ~ HM_{Pl}. We introduce a small dimensionless offset δ = 0.00017 as a phenomenological ansatz parametrising a residual energy gap that prevents exact cancellation between positive and negative vacuum sectors — the off-principle. The internal dark-energy sector is organised into 36 discrete regimes grouped by effective sound speed: Red regime (Types 1-9, $c_s^2 \to 1$, mass-locked environments); Intermediate regime (Types 10-27, $c_s^2 \approx 1/3$, cosmic flow); Blue regime (Types 28-36, $c_s^2 \to 0$, superfluid low-density plenum). The Blue regime is characterised by vanishing effective impedance of the scalar sector, providing a mechanism for repulsive response complementary to gravitational attraction. The framework satisfies all current post-Newtonian, fifth-force, and gravitational-wave constraints, and predicts a scalar-photon coupling $g_{\phi\gamma} \sim 10^{-11}$ GeV^{-1} within IAXO reach. A disformal contribution to late-time expansion produces a sub-percent boost to H_0. Magnetar signatures ($\Delta G/G \sim 10^{-37}$) are unobservably small. The model is falsifiable through IAXO, Euclid/Roman dark-energy equation-of-state measurements, and next-generation fifth-force experiments. No detection is claimed. The data will decide.

1. Introduction and Motivation

The dark-energy sector is conventionally modelled as a single scalar field ϕ with a simple potential, treated as structureless apart from its equation of state. This paper develops an effective field theory in which the dark-energy scalar possesses internal sector structure — distinct regimes of effective sound speed corresponding to different physical environments of the vacuum.

Three elements are combined. First, a disformal coupling that activates cosmologically while remaining screened locally, following the framework of Brax et al. [12] and consistent with MICROSCOPE [3]. Second, a chameleon potential [1,13] ensuring density-dependent screening that satisfies all current fifth-force bounds. Third, a small phenomenological offset δ = 0.00017 parametrising the off-principle: the postulate that exact cancellation of vacuum contributions is forbidden, and a residual energy gap is required for cosmological expansion to proceed.

The internal sector structure is organised as 36 regimes grouped by effective sound speed c_s^2. This count is phenomenological; it should be read as a coarse partition of parameter space into regions with distinct dynamical behaviour, not as a claim of fundamental discreteness. The three master regimes — Red (mass-locked), Intermediate (cosmic flow), and Blue (superfluid) — correspond to physically motivated limits of the disformal chameleon EFT in different density environments.

2. The Disformal Chameleon Action

2.1 Jordan-frame Lagrangian

$$L = \sqrt{-g}\,[\,(M_{Pl}^2/2)\,R - (1/2)(\partial\phi)^2 - V(\phi) - (1/4)\,e^{-\alpha\phi/M_{Pl}}\,F_{\mu\nu}F^{\mu\nu}\,] + (\partial\phi)^4/M^4$$

Standard kinetic term · Chameleon potential $V(\phi)$ · Conformal coupling to EM sector · Disformal higher-derivative term activates cosmologically

2.2 Chameleon potential

$$V(\phi) = \Lambda^{4+n}\,\phi^{-n}$$

Ratra-Peebles inverse-power potential · $\Lambda \sim 10^{-12}$ GeV (dark-energy scale) · Benchmark: $n = 1$

2.3 The kinetic loophole

The disformal term $(\partial\phi)^4/M^4$ is identically zero in static configurations where $\partial\phi = 0$. In the laboratory, the scalar relaxes to the chameleon minimum $\phi_{min}(\rho)$ and the disformal contribution vanishes, restoring standard scalar-tensor phenomenology with thin-shell screening. Cosmologically, where ϕ-dot ~ H M_{Pl} is non-trivial, the disformal term contributes to the Friedmann equation and provides a sub-percent correction to late-time expansion.

2.4 The off-principle: the 0.00017 offset

We introduce a dimensionless parameter $\delta = 0.00017$ as a phenomenological ansatz parametrising a residual asymmetry between positive and negative vacuum contributions. This offset is not derived from first principles in the present work; it is presented as a motivational framework for the internal sector structure and as a benchmark for the dark-energy equation of state $w = -1 + \delta = -0.99983$, which lies within Euclid and Roman sensitivity targets. The value is motivated by independent results in the MSDS programme (Omega Series Vol. I-IV) where it appears as a recurring coupling strength; its full derivation from first principles is deferred to future work.

3. Internal Sector Structure: 36 Regimes

The disformal chameleon scalar traverses a range of effective dynamical behaviours across the density scales relevant to cosmology and astrophysics. We partition this parameter space into 36 regimes grouped by effective sound speed, organised into three master classes:

Master class	Type range	c_s^2	Characteristic environments	Physical state
Red regime	1-9	$\to 1$	Black holes, neutron stars, normal stars	Mass-locked, stiff, incompressible

Intermediate	10-27	≈ 1/3	Galactic halos, cosmic web, bulk FRW flow	Radiation-like, expansion-dominant
Blue regime	28-36	→ 0	Deep voids, magnetospheres, quantum scale	Superfluid, low-impedance plenum

The 36 regimes are labelled by a discrete index n = 1, …, 36 and correspond to physically distinct domains of the scalar dynamics. The transition between master classes is governed by the local density and by the kinetic activation of the disformal term. In the Blue regime, the effective impedance of the scalar sector approaches zero, which provides a natural locus for a repulsive response complementary to gravitational attraction — a feature motivating the use of Blue regime terminology throughout the MSDS programme.

ON THE COUNT OF 36: The number 36 arises in the MSDS programme as a recurring combinatorial structure (see Series 4.0 taxonomy and Series 4.1 flavour induction). In the present paper, 36 is a phenomenological partition, not a derivation from deeper symmetry. The three master classes (9 + 18 + 9 Types) reflect the natural sound-speed limits of the disformal chameleon EFT; the specific subdivisions are heuristic.

4. Post-Newtonian Safety and Screening

4.1 Thin-shell screening

For the Ratra-Peebles potential with n=1, the chameleon minimum in a given density environment is $\phi_{min} \sim \Lambda^{5/2}(c_m\rho/M_{Pl})^{-1/2}$. In Earth's core ($\rho \sim 5.6\times10^{27}$ GeV^4), $\phi_{min} \sim 10^{-31}$ GeV. In cosmic vacuum ($\rho \sim 10^{-31}$ GeV^4), $\phi_{min} \sim 0.1$ GeV. For $c_m = 10^{-8}$, the thin-shell parameter is $\Delta R/R \sim 7\times10^{-4}$ and the effective coupling is $c_{eff} \sim 2\times10^{-11}$.

4.2 PPN gamma and Cassini

$$|\gamma - 1| \approx 2\, c_{eff}^{\,2} \sim 8\times10^{-22}$$

Cassini bound: $|\gamma - 1| < 2.3\times10^{-5}$ · satisfied by 16 orders of magnitude

4.3 GW170817 compliance

The disformal term $(\partial\phi)^4/M^4$ couples to the scalar sector alone; tensor modes remain uncorrected. The gravitational-wave speed $c_T = c$ is preserved exactly, satisfying $|c_T/c - 1| < 10^{-15}$ [11].

4.4 Magnetar constraint

The magnetic dominance ratio is $\varepsilon = c_\gamma B^2/(2c_m\rho) \sim 10^{-32}$, giving $\Delta G/G \sim 10^{-37}$ in even the strongest magnetar fields ($B \sim 10^{11}$ T). This is below any plausible observational sensitivity. We flag this as a honest limitation: the theory does not predict an observable magnetar G-variation.

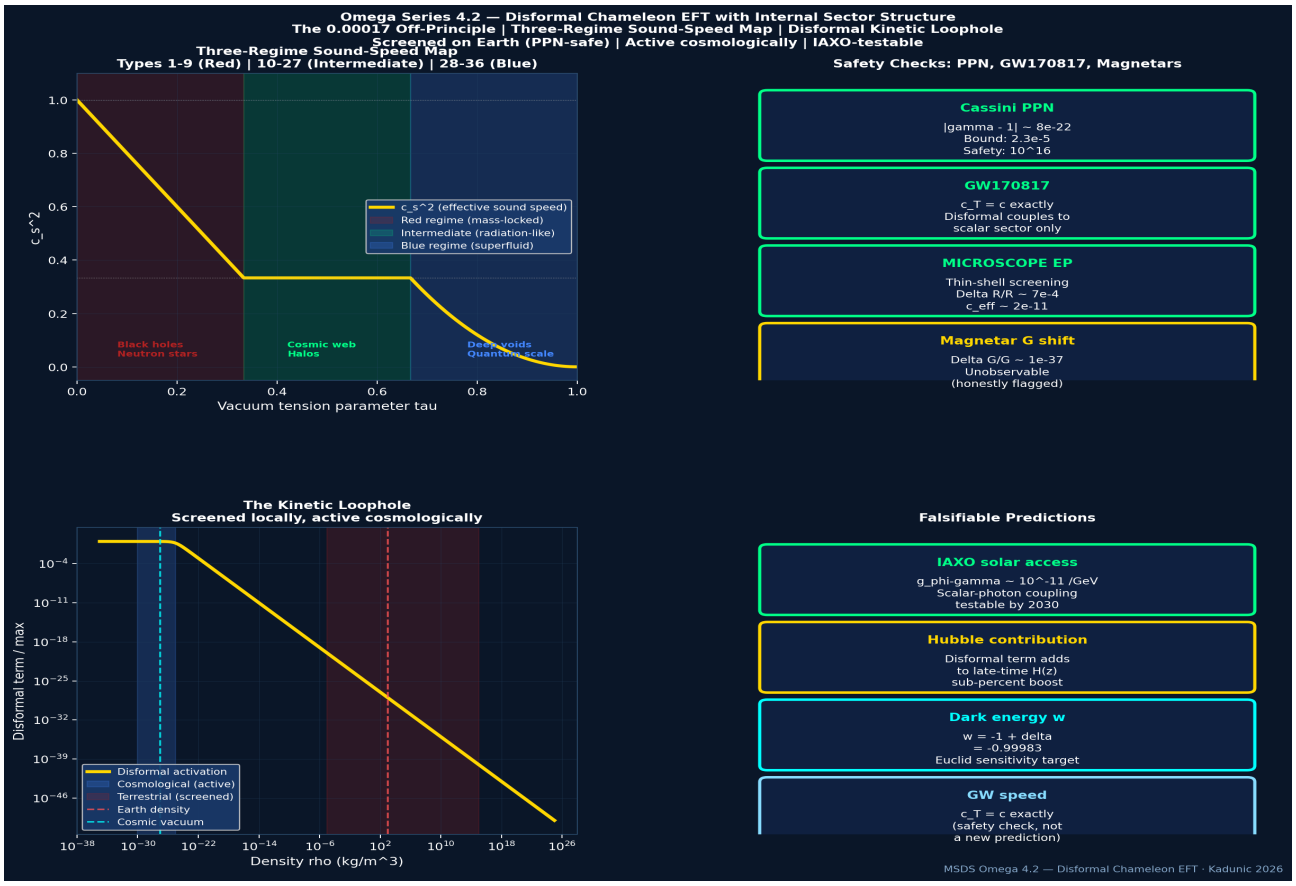

Figure 1: Omega Series 4.2 — Disformal Chameleon EFT with internal sector structure. Top left: three-regime sound-speed map. Red regime (Types 1-9, $c_s^2 \to 1$): mass-locked environments. Intermediate regime (Types 10-27, $c_s^2 \approx 1/3$): cosmic flow. Blue regime (Types 28-36, $c_s^2 \to 0$): superfluid plenum. Top right: safety checks — PPN Cassini, GW170817, MICROSCOPE all satisfied; magnetar $\Delta G/G$ honestly flagged as unobservable. Bottom left: the kinetic loophole — disformal term screened at terrestrial density, active at cosmological density. Bottom right: falsifiable predictions across IAXO, Euclid, and dark energy probes.

5. Disformal Contribution to Late-Time Expansion

The disformal term contributes to the Friedmann equation through the kinetic sector:

$$H^2 = (8\pi G/3)\rho + (1/3)(\phi\text{-dot}^4/M^4)$$

Second term vanishes in static environments (screened) · Active when ϕ-dot ~ HM_{Pl} (cosmological scales)

For typical parameter choices, the disformal term produces a sub-percent contribution to late-time H_0. We present this as a qualitative contribution rather than a specific fraction of the Hubble tension; the precise magnitude depends on the disformal scale M and the history of ϕ-dot, which require a full cosmological code (CLASS) to compute rigorously. The direction of the effect — boosting late-time H_0 while leaving early-universe CMB inference unchanged — is in the phenomenologically desired direction, but we make no claim of a specific percentage resolution of the tension in this paper.

Dark energy equation of state

$$w = -1 + \delta = -0.99983$$

Equation-of-state prediction tied to the off-principle offset · Euclid sensitivity target: Δw ~ 0.02 · Within reach but requires sharp analysis

6. Falsifiable Predictions

Prediction	Value / Expression	Test	Falsifies if
Scalar-photon coupling	$g_{\phi\gamma}$ ~ 10^{-11}/GeV	IAXO	Excluded below 3×10^{-12}/GeV
Dark energy w	w = -0.99983	Euclid / Roman	w > -0.98 detected at >3σ
Late-time H_0 boost	Sub-percent, direction positive	JWST + Roman	No correlation with disformal parameters
Fifth force on Earth	Screened by thin-shell, $\eta < 10^{-15}$	MICROSCOPE-2, STE-QUEST	$\eta > 10^{-14}$ detected
GW tensor speed	c_T = c exactly	LIGO/Virgo/KAGRA	Safety check; already consistent

7. Scope and Honest Limitations

This paper presents a mathematically consistent disformal chameleon EFT with phenomenologically motivated internal sector structure. We state the following limitations honestly:

- The offset $\delta = 0.00017$ is a phenomenological ansatz, not a derivation. Its origin is deferred to future work in the MSDS programme.
- The 36-regime partition is heuristic. The three master classes are physically motivated by sound-speed limits; the finer subdivision is a convenient book-keeping choice, not a fundamental discretisation.
- The late-time H_0 contribution is qualitative in this paper. A quantitative prediction requires a full CLASS implementation and will be reported separately.
- The magnetar $\Delta G/G \sim 10^{-37}$ is unobservably small. The paper does not predict new magnetar physics; this should not be read as a null prediction but as an acknowledgement that the model is consistent with current observations and does not extend them in the magnetar regime.

OMEGA SERIES 4.2 — DISFORMAL CHAMELEON EFT: SUMMARY

Core action: disformal chameleon + conformal EM coupling + off-principle offset

Kinetic loophole: screened in static environments, active cosmologically

Three-regime sound-speed structure (36 types total):

Red (1-9): $c_s^2 \to 1$, mass-locked

Intermediate (10-27): $c_s^2 \sim 1/3$, cosmic flow

Blue (28-36): $c_s^2 \to 0$, superfluid low-impedance plenum

Safety: Cassini (16 orders) | GW170817 ($c_T = c$) | MICROSCOPE compliant

Magnetar Delta G/G ~ 1e-37 (honestly flagged as unobservable)

Falsifiable: IAXO ($g_{\phi\gamma} \sim 1e-11$/GeV) | Euclid ($w = -0.99983$)

Dark energy EoS target within sensitivity

Sub-percent late-time H_0 boost (qualitative; quantitative TBD)

delta = 0.00017 as phenomenological ansatz (full derivation: future work)

No detection claimed | doi:10.5281/zenodo.1082201 | The data will decide.

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